

Siddhartha Reddy Munugala

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SUMMARY

Software Engineer and Founder building AI-powered platforms, real-time web applications, and multi-agent systems. Deployed flagship products (Vibeln, ArenaHQ, Sidewise) under Vibeln LLC, reaching 1,600+ combined registered users with zero paid acquisition. Proficient in Next.js, React, Node.js, Python, Supabase, and RAG/LLM integration. Proven track record of designing microservice architectures, local-first computer vision pipelines, and multi-agent orchestration extensions.

SKILLS

Programming: Python, TypeScript, JavaScript, SQL, C, HTML/CSS
Frameworks: Next.js, React, Node.js, FastAPI, Express, Tailwind CSS
Databases & Storage: PostgreSQL, Supabase, Neon Postgres, Drizzle ORM, Vercel Blob, Redis
AI & Machine Learning: Agentic Workflows (Claude Agent SDK), RAG, embeddings, vector search (pgvector), PyTorch, TensorFlow.js, Tesseract.js OCR, ElevenLabs API, Gemini API, OpenAI API, LangChain
Tools & Infrastructure: Docker, Google Cloud Run, Vercel, Render, Git, GitHub Actions (CI/CD), WebRTC, WebSockets, REST APIs

EXPERIENCE

Founder and Lead Engineer — Vibeln LLC

Jan 2025 - Present

vibein.us / arenahq.us / sidewise.vercel.app

- Architected ArenaHQ (arenahq.us), a live 1v1 coding sandbox using Next.js 15, Supabase, and CodeMirror 6, integrating WebRTC peer-to-peer streams (RTCPeerConnection) and a sandboxed Piston execution API (800+ users).
- Launched Vibeln (vibein.us), an AI-powered social matchmaking campus network (821 registered users, 40% DAU), implementing a custom matchmaking ELO rating algorithm and realtime WebSockets.
- Engineered Sidewise's RAG research assistant utilizing Brave Search API and Gemini 2.5 Flash, separating retrieval from synthesis to guarantee 100 percent source-backed summaries with zero model hallucinations.
- Diagnosed and resolved 8+ production-critical database and auth pipeline failures under live user load, maintaining 99.9% uptime.

IT Coordinator and Publicity Chair — NSLS Executive Board

Sep 2024 - Present

University of South Dakota

- Automated membership tracking and recruitment pipelines using Python and Google Apps Script, reducing onboarding processing overhead by 50 percent (from 4 weeks to 2 weeks).

PROJECTS

Scout — Real-Time AI Vision Assistant for the Visually Impaired

2026

- Built a three-tier local-first accessibility vision pipeline (scoutboy.vercel.app) routing frames between TensorFlow.js COCO-SSD (80-100ms) and serverless Gemini 2.5 Flash with a 10s abort ceiling.
- Designed a frame-differencing canvas filter (12% threshold) to skip redundant AI inferences; offloaded rendering to Web Workers and resolved iOS Safari TTS limits via a keepalive watchdog.

eko — AI-Assisted Multi-Track DAW and Audio Platform

2026 - Present

- Engineered FastAPI audio separation services on Google Cloud Run performing pitch/rhythm analysis (librosa), MIDI synthesis, and ElevenLabs voice cloning.
- Developed a recommendation engine using PyTorch autoencoders to compress 70-D audio vectors to 16-D embeddings, integrated with a PennyLane variational quantum circuit (4 qubits) and pgvector search.

prism — CTO & Multi-Agent VS Code Extension

2026

- Built a VS Code extension implementing a planner/executor multi-agent system wrapping the Claude Agent SDK as an embedded library.
- Engineered a capability-based security middleware proxy to intercept and halt privileged shell executions and filesystem writes for explicit user-in-the-loop approval.

KUSHI & ChallengeTrade — Connection & Skill-Based Commerce Engines

2025 - 2026

- Designed KUSHI, a slow connection platform built around emotional compatibility, replacing swiping mechanics with anonymous letter writing and an AI trust assistant to pace conversation depth.
- Architected ChallengeTrade, a skill-based competitive marketplace where physical items are won through skill challenges (e.g. esports), incorporating prediction pools for live spectator interaction.

EDUCATION

B.A. Computer Science, Minor in Innovation and Entrepreneurship

Expected May 2027

University of South Dakota • GPA: 3.0/4.0

Coursework: Data Structures & Algorithms, Systems Programming, Database Systems, Software Engineering

NSF I-Corps Customer Discovery Program Graduate • NSLS National Leadership Inductee